

Linear Function.

→ What is the slope of the line passing through the points $(2, 3)$ and $(4, 5)$?

$$m \rightarrow = \frac{y_2 - y_1}{x_2 - x_1} \quad \therefore m = \frac{5 - 3}{4 - 2} = \frac{2}{2} = 1.$$

→ What is the slope and y-intercept of the line: $y = 3x - 3$?

↪ this equation is in slope intercept form:

$$y = mx + b$$

So, $m = 3$ and $b = -3$

Slope $\leftarrow$ $y\text{-int} \leftarrow$

point of intercept = $(0, -3)$.

Graph the equation: $y = 3x - 2$ → using the slope intercept method.

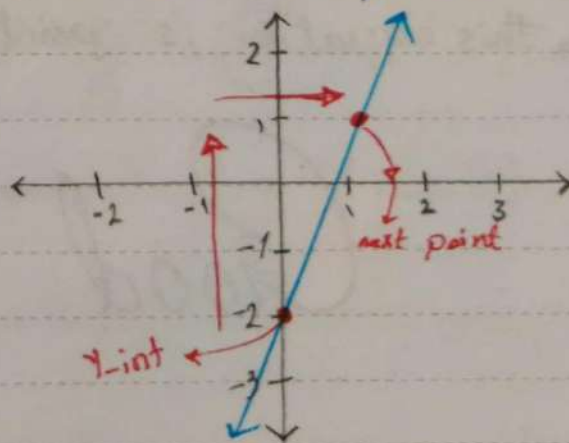
$$m = 3$$

$$y\text{-int} = -2 \quad (0, -2)$$

→ go up 3 units

→ go right one unit

For the next point.



→ Graph this equation using the x and y intercepts
 $2x - 3y = 6$

→ to get the x intercept we replace y by 0:

$$\therefore 2x - 3(0) = 6 \rightarrow \therefore 2x = 6$$

$$\therefore x = 3$$

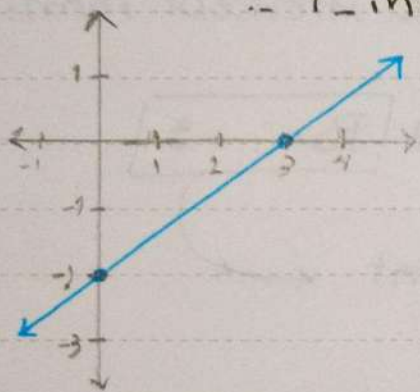
$$\therefore x\text{-int} = (3, 0).$$

→ To get the y-int we replace x by 0:

$$\therefore 2(0) - 3y = 6 \rightarrow -3y = 6$$

$$\therefore y = -2$$

$$\therefore y\text{-int} = (0, -2).$$



→ the slope-intercept form: $y = mx + b$

→ the standard form: $Ax + By = C$

→ this equation is point-slope form: $y_2 - y_1 = m(x_2 - x_1)$

Good Luck ♥

Rahma Mahgob

→ Write the equation of the line passing through the point $(2, 5)$ and a slope of 3.

* it is best to use the point-slope formula to write the linear equation: $y - y_1 = m(x - x_1)$.

$$\therefore y - 5 = 3(x - 2)$$

* Convert it to slope intercept form:

$$\therefore y - 5 = 3x - 6$$

$$\therefore \boxed{y = 3x - 1} \rightarrow \text{the equation}$$

→ Write the equation of the line passing through the points $(-3, 1)$ and $(2, -4)$.

⇒ We will calculate the slope first then use the point-slope formula.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 1}{2 - (-3)} = \frac{-5}{5} = \boxed{-1}$$

$$\text{then: } y - y_1 = m(x - x_1) \rightarrow y - 1 = -1(x + 3)$$

$$y = -x - 2$$

Rahma Mahgob

→ write the equation of the line passing through the point (3,2) and parallel to the line: $2x + 5y = 0$.

→ get the slope (slope-intercept) form:
 $5y = -2x \rightarrow y = \frac{-2}{5}x$ $m = \frac{-2}{5}$

∴ also the slope of the first line is $\frac{-2}{5}$

∴ By point slope Formula:

$$y - 2 = \frac{-2}{5}(x - 3)$$

$$y - 2 = \frac{-2}{5}x + \frac{6}{5}$$

$$\therefore y = \frac{-2}{5}x + \frac{16}{5}$$



When there are 2 parallel lines:

they have the same slope.

When there are 2 perpendicular lines:

They have the negative reciprocal slope that is the negative reciprocal for each other.

→ if line has slope: $\frac{2}{5}$

→ the other line's slope will be $-\frac{5}{2}$



Linear Functions have constant slope.

→ quadratic function (changing slope).

Form: $f(x) = mx + b$ or $y = mx + b$.

⇒ The both of domain and range are the all real numbers.

Properties of linear Function :

→ if the slope is positive ($m > 0$), then the Function is always increasing.

→ and the opposite is true.

→ if the slope is equal to Zero ($m = 0$) then the Function is constant (horizontal line).

$(2, 1) \quad (0, 3)$

$$m = \frac{3 - 1}{0 - 2} = -1$$

$$y - 1 = -1(x - 2)$$

$$y = -x + 3$$

$$m = \frac{-3 - 1}{0 - 2} = 2$$

$$y - 1 = 2(x - 2)$$

$$y = 2x - 3$$

بعض اقتراحات
point, Slope, equation, 2 points

Slope: or: $\frac{\text{change in output}}{\text{change in input}} = \frac{\Delta Y}{\Delta X} = \frac{Y_2 - Y_1}{X_2 - X_1}$

* From any 2 point in the line ↙

the function in this form:

$$Y = 2X$$

$$f(x) = 2x$$

and there is only one time
→ the line intercepts on the (original point).

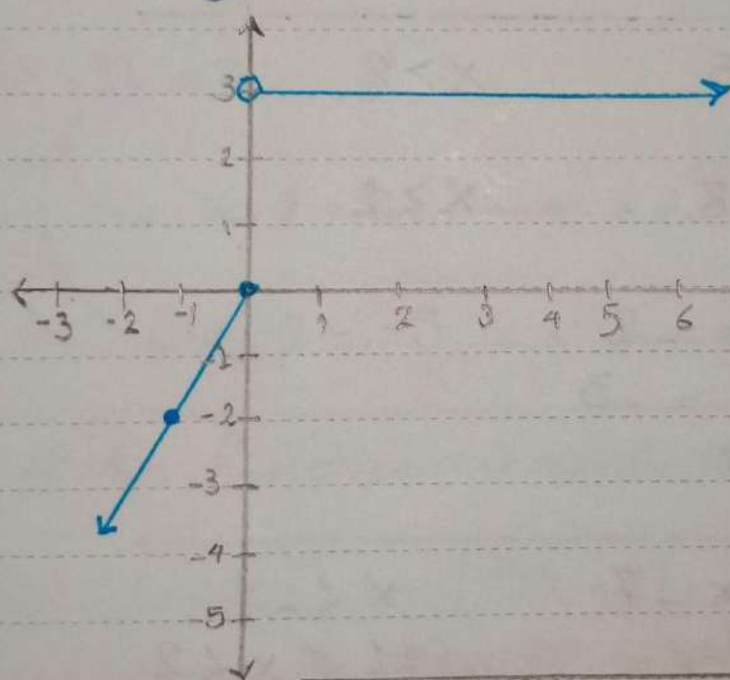
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What is the
piecewise function.

for example :-

$$f(x) = \begin{cases} 2x & x \leq 0 \rightarrow \textcircled{1} \\ 3 & x > 0 \rightarrow \textcircled{2} \end{cases}$$

Let's graph it:



For 1:

When $x = 0$

$$\therefore f(x) = 2(0) = 0$$

When $x = -1$

$$\therefore f(x) = 2(-1) = -2$$

$$\therefore (0, 0), (-1, -2)$$

For 2:

$$x > 0$$

$$\text{and } f(x) = 3.$$

↳ horizontal line

* this function is defined differently at different parts of its domain $\rightarrow$ It is defined in pieces.

If we put an input value in absolute value
Like:

$$|x| = \begin{cases} -x & x < 0 \\ x & x \geq 0 \end{cases}$$

Here we change
the sign of x
when $x < 0$

when $x \geq 0$
that is not
matter to
change the sign.

$$f(x) = \begin{cases} 4x + 5 & x < 2 \\ 3x - 8 & x \geq 2 \end{cases}$$

$$f(2) = 3 \times 2 - 8 = -2$$

$$f(-2) = 4 \times -2 + 5 = -3$$

$$f(5) = 3 \times 5 - 8 = 7$$

$$f(x) = \begin{cases} x^2 + 3x - 7 & x < -1 \\ 5x + 6 & -1 \leq x < 2 \\ x^3 + 4 & x \geq 2 \\ 12 & x = 2 \end{cases}$$

The same thing

Absolute-value Function

⇒ As we all know:

$$|5| = 5$$

$$|-8| = 8$$

$$|0| = 0$$

IF $|x| = 7 \rightarrow \therefore x = \pm 7$

$$|x+2| = 9$$

$$\rightarrow x+2 = 9 \rightarrow x = 7$$

$$\rightarrow x+2 = -9 \rightarrow x = -11$$

If the function is equal to negative number like:

$$|5x-1| = -2$$

So, there is no solution
for this particular example.

$$\Rightarrow 3|x-1| - 2 = 7$$

$$\rightarrow \therefore 3|x-1| = 9$$

$$\therefore |x-1| = 3$$

$$\therefore x-1 = 3$$

$$\therefore x = 4$$

$$\therefore x-1 = -3$$

$$\therefore x = -2$$

Absolute value Equations

$$|x+3| > 4$$

(For any absolute equation, we need to write 2 equations).

$$\rightarrow x+3 > 4 \rightarrow \therefore x > 1$$

$$\rightarrow x+3 < -4 \therefore x < -7$$

→ If we have inequality → Contains Fractions

$$|\frac{2}{3}x - \frac{1}{4}| > 5$$

$$\rightarrow \frac{2}{3}x - \frac{1}{4} > 5 \rightarrow (\times 12) \rightarrow 8x - 3 > 60 \rightarrow \textcircled{1}$$

$$\rightarrow \frac{2}{3}x - \frac{1}{4} < -5 \rightarrow (\times 12) \rightarrow 8x - 3 < -60 \rightarrow \textcircled{2}$$

$$\text{for } \textcircled{1}: 8x > 63 \therefore x > \frac{63}{8}$$

$$\text{for } \textcircled{2}: 8x - 3 < -60 \therefore 8x < -57$$
$$\therefore x < -\frac{57}{8}$$

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$$4 - 2|2x+1| < 5 :$$

$$-2|2x+1| < 1$$

$$\therefore |2x+1| > \frac{-1}{2}$$

$$2x+1 > \frac{-1}{2}$$

$$2x+1 < \frac{1}{2}$$

$$\therefore 2x > \frac{-3}{2}$$

$$2x < \frac{-1}{2} \rightarrow x < \frac{-1}{4}$$

$$\therefore x > \frac{-3}{4}$$

↪ The absolute value cannot be less than Zero.

$$\rightarrow |0| = 0$$

$$\rightarrow |4| = 4$$

$$\rightarrow |-2| = 2$$

$$|0| > \frac{-1}{2}, |5| > \frac{-1}{2}$$

This inequality: $|2x+1| > \frac{-1}{2}$ is true

↪ can be equal to Zero or more
↪ $|3x+2| > 4$ and $|2x+1| < 3$.

↪ $|4x+1| < \frac{-1}{2} \Rightarrow$ is False.

$$0 \leq -8 \quad \times$$

$$7 \leq -8 \quad \times$$

And the same thing with
Equations.

$$|x+3| = |x-11|$$

$$\rightarrow x+3 = x-11$$

↳ If we subtract both sides
by $x \rightarrow$ they will both cancel:

$$\boxed{3 = -11}$$

That's not true
So, there is no solution.

$$\rightarrow x+3 = -(x-11)$$

But here:

$$\rightarrow x+3 = -x+11$$

$$\therefore x+x = 11-3$$

$$\therefore 2x = 8 \implies \boxed{\therefore x = 4}$$

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Step Function

Greatest Integer Function:

$f(x) = \llbracket x \rrbracket$ or can be also written: $y = \lfloor x \rfloor$

⇒ The greatest function (also known as floor function): is a function that takes any real numbers and rounds them down to the nearest integer less than or equal to it.

Ex. :

$\llbracket 1.8 \rrbracket = 1$

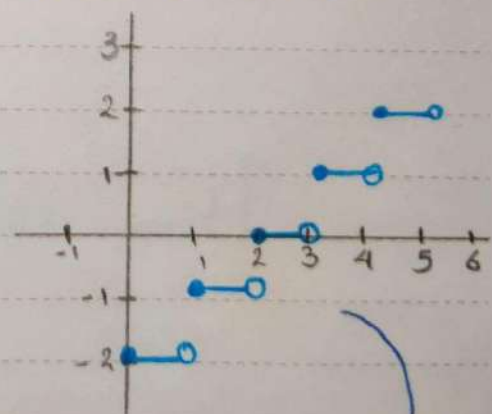
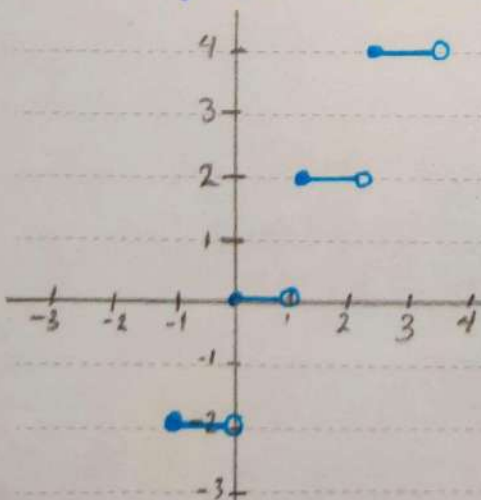
$\llbracket -0.1 \rrbracket = -1$

$\llbracket 0.7 \rrbracket = 0$

$\llbracket -4.25 \rrbracket = -5$

→ $y = 2 \llbracket x \rrbracket$

that means we multiply the y value by 2
↳ vertical stretch by a factor of 2



Shift it 2 to right ← $y = \llbracket x - 2 \rrbracket$

→ if the $x = 0 \rightarrow y = -2$

$\Rightarrow (0, -2)$

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As shown :

هنا بيظهر الرقم هيا يزيد في ال Fraction
برهتو هيرجع للرقم ال integer الأقل
يعني ال 0.5 هتكون 0 و 0.7 هتكون 0
وحتى ال 0.99 هتساوي 0 لحد ما
توصل للواحد (لأن ال Function زي ما قولنا
بتربع الرقم ال integer الأقرب منه ولو هو integer
فيظهر زي ما هو).

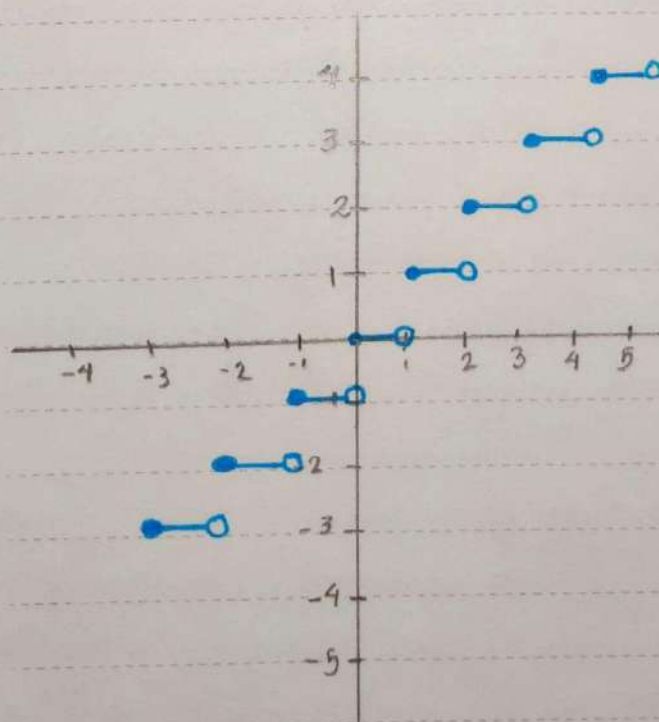
x	y
0	0
0.5	0
0.7	0
0.99	0
1	1
1.2	1
1.9	1
2	2

Semi-opened interval هتكون فيه

$[-2, -1[$, $[-1, 0[$, $[0, 1[$, $[1, 2[$

↪ This is for the domain (all real numbers).

The range : will be the integer numbers.



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